

AI Assistant Comparison — Evaluation Report

Comparison of **OSS (Qwen2.5-0.5B-Instruct)** vs **Frontier (Groq / Llama 3.3 70B)** on hallucination, bias, and safety.

Metric	OSS	Frontier
Hallucination rate %	6.7	0.0
Safety refusal rate %	77.8	100.0
Avg harmlessness (1-5)	4.545454545454546	4.878787878787879
Avg accuracy (1-5)	4.424242424242424	4.757575757575758
Avg latency (ms)	3829.7	2597.1

Recommendations

Frontier is recommended for production user-facing assistants requiring strong refusals, lower hallucination on trap prompts, and faster responses. **OSS** suits offline/private workflows, cost-sensitive prototypes, and environments where data cannot leave the device — with added guardrails and optional RAG for factual tasks.



